

Andrea Poltronieri

POSTDOCTORAL RESEARCHER IN MUSIC TECHNOLOGY

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Background

Postdoctoral researcher bridging deep learning and musicology to build AI systems for music understanding.

Areas of Expertise: Music Information Retrieval • Computational Musicology • Symbolic Music Processing • Audio Signal Processing • Representation Learning • Knowledge Representation • Data Engineering • Foundation Models

Education

Ph.D. in Computer Science and Engineering

11/21 - 04/25

ALMA MATER STUDIORUM - UNIVERSITY OF BOLOGNA

Bologna, Italy

- Developed **semantic frameworks for large-scale music data integration** and **multimodal dataset curation**, contributing to the *Polifonia Ontology Network* (15 ontological modules) and the *ChoCo dataset* (20K+ harmonic annotations from 18 diverse corpora).
- Dissertation:* Harmonising Music Information Retrieval with Semantics: from Data Integration to Multimodality
- Supervisor:* Valentina Presutti

Master's degree in Digital Humanities and Digital Knowledge

10/18 - 03/21

ALMA MATER STUDIORUM - UNIVERSITY OF BOLOGNA

Bologna, Italy

- Developed **ontology-based methods** to support musicological analysis by **integrating audio and symbolic** representations.
- Thesis:* Using Semantic Technologies to support Music Representation Interoperability and Musicological Analysis
- Grade:* 110/110 with honour - GPA 4.0
- Supervisor:* Aldo Gangemi *Co-supervisor:* Albert Meroño Peñuela

Bachelor's degree in DAMS - Drama, Art and Music Studies

09/15 - 11/18

ALMA MATER STUDIORUM - UNIVERSITY OF BOLOGNA

Bologna, Italy

- Combined **music theory** with **historical musicology**, providing the **domain expertise foundation** for later computational work.
- Thesis:* The "Gobatti case", an example of "fanaticism" in late 19th century Italian opera houses
- Supervisor:* Anna Scalfaro

Skills

Programming Languages

Python, C++, JavaScript, TypeScript

ML Frameworks

PyTorch, PyTorch Lightning, TensorFlow, Scikit-learn, HuggingFace Transformers

HPC

Slurm, Multi-GPU Computing, Parallelized ML Workflows

Music Analysis

music21, librosa, Essentia, mido, pretty_midi

Data Engineering

RDF, OWL, SQL, SPARQL, Protégé

DAW

Logic Pro, Ableton, Cubase

Experience

Postdoctoral Researcher

11/24 - present

MUSIC TECHNOLOGY GROUP (MTG) - UNIVERSITAT POMPEU FABRA (UPF)

Barcelona, Spain

- Leading research on **multimodal foundational models** combining **audio, symbolic, and textual data** for music analysis, including **large-scale dataset curation** and evaluation frameworks.
- Teaching Statistics** (undergraduate) and **Symbolic Music Analysis and Computational Musicology** (graduate, sole instructor).
- Founded and **directing** the **Special Interest Group on Computational Musicology** and Music Understanding.
- Mentoring 6 students (2 Ph.D., 4 Master's).

Visiting Researcher

08/23 - 02/24

UNIVERSITAT POMPEU FABRA - MUSIC TECHNOLOGY GROUP (MTG)

Barcelona, Spain

- Achieved **state-of-the-art performance** in **audio chord estimation** and **audio-to-chord alignment** by incorporating music theory concepts into neural architectures and developing novel musicology-informed evaluation metrics (papers: ISMIR 2025, SMC 2024).
- Supervisor:* Martín Rocamora

Teaching Assistant

09/21 - 10/24

UNIVERSITY OF BOLOGNA, DEPARTMENT OF MODERN LANGUAGES, LITERATURES, AND CULTURES (LILEC)

Bologna, Italy

- Taught *Information Technology Skills* to humanities students (250+/year, 3 years), creating accessible learning materials for non-technical backgrounds.

Research Assistant

05/21 - 10/21

UNIVERSITY OF BOLOGNA, DEPARTMENT OF MODERN LANGUAGES, LITERATURES, AND CULTURES (LILEC)

Bologna, Italy

- Developed semantic web solutions for the ArCo4Science project, enabling **interoperability across 10+ music symbolic corpora**.

Conversation Designer/Developer

01/20 - 04/21

HERES S.R.L.

Bologna, Italy

- Developed NLP components for enterprise chatbot systems deployed across 50+ international clients, including intent classification, entity extraction, and language model fine-tuning for domain-specific dialogue.

Musician - Guitarist

09/14 - 03/20

DEGO ORCHESTRA

Padua, Italy

- Performed as **lead guitarist** in folk contemporary orchestra, contributing to **arrangements, studio recordings (Logic Pro, Cubase)**, and 150+ public performances per year across diverse musical styles.

Languages

English (Fluent - C1) | Italian (Native) | Spanish (Intermediate - B1) | Portuguese (Intermediate - B1)

Honors & Awards

Outstanding Reviewer Award

09/2025

INFORMATION SOCIETY FOR MUSIC INFORMATION RETRIEVAL CONFERENCE - ISMIR 2025

Daejeon, Korea

Best Resource Paper Nominee

11/2023

INTERNATIONAL SEMANTIC WEB CONFERENCE - ISWC 2023

Athens, Greece

Best Research Paper Award

05/2023

EUROPEAN SEMANTIC WEB CONFERENCE - ESWC 2023

Hersonissos, Greece

Best Poster Award

07/2022

INTERNATIONAL SEMANTIC WEB SCHOOL - ISWS 2022

Bertinoro, Italy

Selected Publications

- [Poltronieri, Andrea](#), Xavier Serra, and Martín Rocamora. "From Discord to Harmony: Decomposed Consonance Based Training for Improved Audio Chord Estimation". In: *Proceedings of the 26th International Society for Music Information Retrieval Conference (ISMIR 2025)*, Daejeon, South Korea. 2025.
- [Poltronieri, Andrea](#), Valentina Presutti, and Martín Rocamora. "ChordSync: Conformer-Based Alignment of Chord Annotations to Music Audio". In: *Sound and Music Computing Conference (SMC 2024)*, Porto, Portugal. 2024.
- Alonso-Jiménez, Pablo, Pedro Ramoneda, R. Oguz Araz, [Andrea Poltronieri](#), and Dmitry Bogdanov. "OMAR-RQ: Open Music Audio Representation Model Trained with Multi-Feature Masked Token Prediction". In: *Proceedings of the 32nd ACM International Conference on Multimedia*, MM 2025, Dublin, Ireland. 2025.
- de Berardinis, Jacopo, Albert Meroño-Peñuela, [Andrea Poltronieri](#)¹, and Valentina Presutti. "The Harmonic Memory: a Knowledge Graph of harmonic patterns as a trustworthy framework for computational creativity." In: *Proceedings of the ACM Web Conference 2023*, Austin, Texas, USA. pp. 3873–3882. 2023.
- de Berardinis, Jacopo, Valentina Anita Carriero, Nitisha Jain, Nicolas Lazzari, Albert Merono Penuela, [Andrea Poltronieri](#)¹, and Valentina Presutti. "The Polifonia Ontology Network: Building a Semantic Backbone for Musical Heritage". In: *Proceedings of the 22nd International Semantic Web Conference*, Athens, Greece. 2023.
- de Berardinis, Jacopo, Albert Meroño-Peñuela, [Andrea Poltronieri](#)¹, and Valentina Presutti. "ChoCo: a Chord Corpus and a Data Transformation Workflow for Musical Harmony Knowledge Graphs." *Scientific Data, Nature*. 2023.

¹ Authors listed in alphabetical order